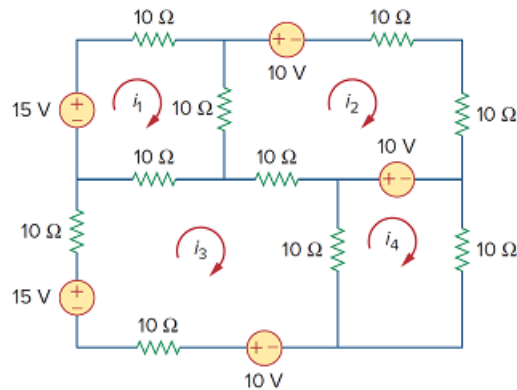


Problem

Write the mesh-current equations for the circuit in Fig. 3.117.

Figure 3.117:



Step-by-step solution

Step 1 of 12

Refer to the circuit in Figure 3.117 in the textbook.

If a circuit has N meshes, the mesh-current equations can be expressed in terms of resistances:

$$\begin{pmatrix} R_{11} & \cdots & R_{1N} \\ \vdots & \ddots & \vdots \\ R_{N1} & \cdots & R_{NN} \end{pmatrix} \begin{pmatrix} i_1 \\ \vdots \\ i_N \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix} \text{ or simply } \mathbf{R}\mathbf{i} = \mathbf{v}.$$

Where,

Sum of the resistances in mesh k is R_{kk} ,

Negative of the sum of the resistances in common with meshes k and j , $k \neq j$ is $R_{kj} = R_{jk}$

Unknown mesh current for mesh k in the clockwise direction is i_k

Sum taken clockwise of all independent voltage sources in mesh k , with voltage rise treated as positive is v_k .

Step 2 of 12

The circuit in Figure 1 has five meshes, so we need five mesh equations. This implies that the size of the resistance matrix $\mathbf{R}$, is 4 by 4.

Each of the diagonal terms is the sum of the resistances in the related mesh, while each of the off-diagonal terms is the negative of the resistances common to meshes.

Calculate the diagonal terms of $\mathbf{R}$, in ohms:

Calculate the value of R_{11} .

$$\begin{aligned} R_{11} &= 10 + 10 + 10 \\ &= 30 \end{aligned}$$

Therefore, the value of the R_{11} is 30.

Step 3 of 12

Calculate the value of R_{22} .

$$\begin{aligned} R_{22} &= 10 + 10 + 10 + 10 \\ &= 40 \end{aligned}$$

Therefore, the value of the R_{22} is 40.

Step 4 of 12

Calculate the value of R_{33} .

$$\begin{aligned} R_{33} &= 10 + 10 + 10 + 10 + 10 \\ &= 50 \end{aligned}$$

Therefore, the value of the R_{33} is 50.

Step 5 of 12

Calculate the value of R_{44} .

$$\begin{aligned} R_{44} &= 10 + 10 \\ &= 20 \end{aligned}$$

Therefore, the value of the R_{44} is 20.

Step 6 of 12

Calculation of non-diagonal terms:

Calculate the value of R_{12} .

$$R_{12} = -10$$

Therefore, the value of the R_{12} is -10.

Since R_{21} is equal to R_{12} ,

$$R_{21} = -10$$

The value of the R_{21} is -10.

Step 7 of 12

Calculate the value of R_{13} .

$$R_{13} = -10$$

Therefore, the value of the R_{13} is -10.

Since R_{31} is equal to R_{13} ,

$$R_{31} = -10$$

The value of the R_{31} is equal to -10.

Step 8 of 12

Calculate the value of R_{23} .

$$R_{23} = -10$$

Therefore, the value of the R_{23} is -10 .

Since R_{23} is equal to R_{32} ,

$$R_{32} = -10$$

The value of the R_{23} is equal to -10 .

Step 9 of 12

Calculate the value of R_{24} .

$$R_{24} = 0$$

Therefore, the value of the R_{24} is 0 .

Since R_{42} is equal to R_{24} ,

$$R_{42} = 0$$

The value of the R_{42} is equal to 0 .

Step 10 of 12

Calculate the value of R_{34} .

$$R_{34} = -10$$

Therefore, the value of the R_{34} is -10 .

Since R_{43} is equal to R_{34} ,

$$R_{43} = -10$$

The value of the R_{43} is equal to -10 .

Step 11 of 12

Calculate the value of R_{14} .

$$R_{14} = 0$$

Therefore, the value of the R_{14} is 0 .

Since R_{41} is equal to R_{14} ,

$$R_{41} = 0$$

The value of the R_{41} is 0 .

Step 12 of 12

Write node voltage equations in matrix representation.

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ R_{31} & R_{32} & R_{33} & R_{34} \\ R_{41} & R_{42} & R_{43} & R_{44} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

Substitute -10 for R_{13} and R_{31} , -10 for R_{23} and R_{32} , 0 for R_{41} and R_{14} , -10 for R_{43} and R_{34} , -10 for R_{21} and R_{12} , 0 for R_{42} and R_{24} , 30 for R_{11} , 40 for R_{22} , 50 for R_{33} , 20 for R_{44} , 15 for v_1 , $-10+10$ for v_2 , $15+10$ for v_3 and -10 for v_4 .

$$\begin{bmatrix} 30 & -10 & -10 & 0 \\ -10 & 40 & -10 & 0 \\ -10 & -10 & 50 & -10 \\ 0 & 0 & -10 & 20 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 15 \\ -10+10 \\ 15+10 \\ -10 \end{bmatrix}$$

$$\begin{bmatrix} 30 & -10 & -10 & 0 \\ -10 & 40 & -10 & 0 \\ -10 & -10 & 50 & -10 \\ 0 & 0 & -10 & 20 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 15 \\ 0 \\ 25 \\ -10 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -2 & -2 & 0 \\ -2 & 8 & -2 & 0 \\ -2 & -2 & 10 & -2 \\ 0 & 0 & -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 5 \\ -2 \end{bmatrix}$$

Therefore, the set of mesh current equations for the given circuit is,

$$\begin{bmatrix} 6 & -2 & -2 & 0 \\ -2 & 8 & -2 & 0 \\ -2 & -2 & 10 & -2 \\ 0 & 0 & -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 5 \\ -2 \end{bmatrix}$$